PROJECT GUIDE

Islet

A Dynamic-Island-style notch for Windows 11

Music, Claude Code and Codex activity, notifications, battery and more,
in one small black pill at the top of your screen.

MADE BY	VERSION	PLATFORM	BUILT WITH
Gopinath	**2.0.0**	**Windows 10 / 11**	**Electron 37**

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SECTION 01

Why Islet exists

Phones and Macs keep live status in one place at the top of the screen. Windows spreads it across the taskbar, the tray, a corner of banners and hidden terminal windows. Islet pulls it into one pill.

One glance

See what is playing, what your AI coding agent is doing, and new messages without switching windows.

Act in place

Skip songs, seek, change volume, and Allow or Deny Claude Code permission requests from the notch.

Out of the way

Click-through except over the notch, hides in fullscreen apps, and goes to a single dot when idle.

Local and light

No account, no cloud, no tracking, and no unsigned helper exe. Works with Smart App Control on.

The real thing



Open notch: agent sessions on the left, Spotify in the middle, time and battery on the right, tools at the edge.

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SECTION 02

Every feature

Grouped by area. Everything below ships in this repository today.

Music

- Spotify, or any player using Windows media controls (YouTube in a browser, VLC, ...)
- Album art, title, artist; accent colour taken from the art; art clears when a track has none
- Previous / play-pause / next; click the bar to seek
- Scroll over the music side to change Spotify's own volume
- Like the song (Spotify's Alt+Shift+B shortcut)
- Live wave from Spotify's audio only (24 FFT bands); stops drawing once the bars settle
- Timed lyrics (lrclib.net) under the seek bar
- Enlarged album view over a blurred backdrop
- Song-change peek; optional Spotify Web API (Premium)

Claude Code and Codex

- Sessions from both agents in one column, newest first (up to five; waiting sessions are always kept)
- Status per session: working, needs you, done
- Last 3 actions (prompt, tools, edited files)
- Allow / Deny Claude Code permission requests from the notch
- Session chips to switch project; Codex chips have a blue dot
- One-click hook install with backups; untick removes only Islet's handlers
- Falls back to reading transcripts when hooks are off

Notifications

- Every Windows toast becomes a notch pop-up NEW with the full message: 400 px wide, grows up to about 7 lines
- Display time grows with message length (4.5 s + 40 ms per character, max 12 s)
- Real app logos; click to open the app; hover to keep it
- Windows' own banners off while running, restored on quit NEW even after a crash
- NEW On the lock screen Windows' banners come back, so messages still show there
- NEW Hidden pop-ups wait, then show (latest 5)
- Do Not Disturb, with an optional daily schedule

Battery and system

- Battery level and charging state
- NEW Black hole animation when the charger goes in (~2.8 s, "CHARGING INITIATED") or out (~1.8 s, "RUNNING ON BATTERY"); Cinematic or Subtle style
- Charging: pulsing green edge glow and a streak on the bottom edge (paused while the black hole plays)
- One-time low-battery warning
- Mic, camera and screen-capture dots while in use
- Clock and date column; hides behind fullscreen apps

Info

- Weather by city (Open-Meteo, no key), C or F
- Next calendar event from a private ICS link; cancelled events, EXDATEs and moved occurrences handled NEW follows each event's time zone
- Browser download progress and a finished pop-up
- Timer from the tray, shown as a chip

Tools and window

- Clipboard history: last 5 texts, memory only
- File shelf: drop files in, drag them out (no duplicates)
- Hover to open, click elsewhere to close
- Ctrl+Alt+N: pill, open, hidden; Ctrl+Alt+M: next monitor
- NEW Alt+F4 can't close the notch by mistake; use tray > Quit
- Settings window, tray menu, start with Windows, single instance

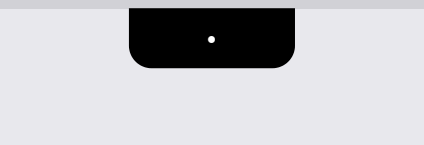
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SECTION 03

How it looks: every state

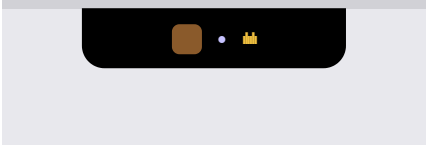
The notch is always black and hangs from the top edge of the screen. It changes size and content with what is going on.

Closed pill



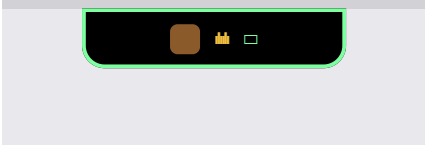
Quiet

Nothing playing, not charging, agent idle: one dot.



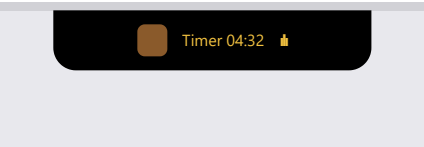
Playing

Album art, Claude dot (lit when active), live wave.



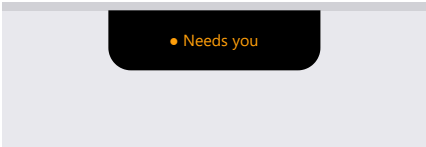
Charging

Green edge glow pulses; a light streak runs along the bottom.



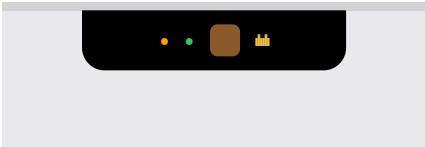
Timer / meeting / download

A chip shows the most urgent item.



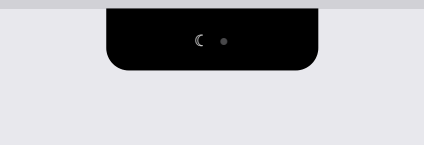
Agent needs you

Amber pulsing dot while a permission waits.



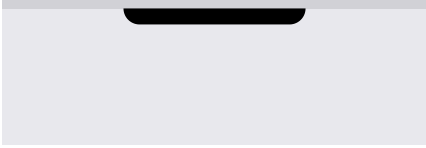
Mic / camera in use

Orange and green dots on the left.



Do Not Disturb

Moon icon; pop-ups are held back.



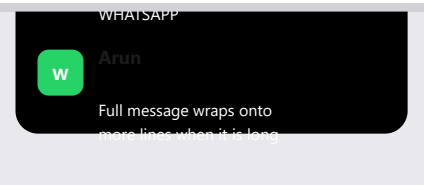
Idle line

Optional (idleHide): a thin 7 px line.

Hidden

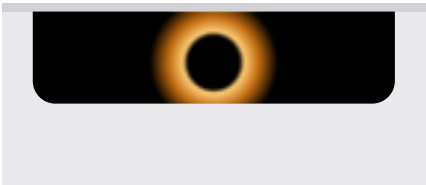
Ctrl+Alt+N or a fullscreen app: nothing on screen.

Pop-ups and open



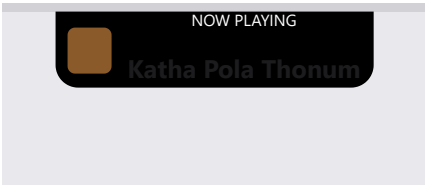
Notification pop-up NEW

400 px wide, height grows with the text.



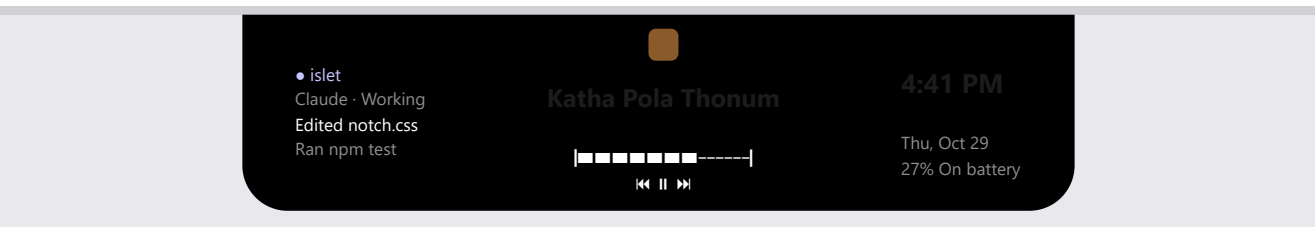
Black hole NEW

340 x 84 px, plays on charger in / out.



Song change peek

Opens briefly on a new song.



Open

Hover: 640 x 150 px. Agents, music, time and battery, tools. Click elsewhere to close. The green glow follows the edge while charging.

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Real screenshots



Pill



Pop-up



Album view

SECTION 04

Tech stack

Small and native to Windows. No UI framework, no database server, no cloud.

Layer	Technology	Used for
App shell	Electron 37 (Chromium + Node.js)	Frameless, transparent, always-on-top window; tray; global hotkeys; powerMonitor; IPC
UI	HTML, CSS, vanilla JavaScript	The notch (renderer/) and the settings window (settings/)
Animation	Canvas 2D	Black hole charger animation (renderer/blackhole.js); runs only while playing
Main process	Node.js 20+ (CommonJS)	Window control, config, timers, HTTP bridge, spawning helpers
Windows access	Windows PowerShell 5.1	Media session, privacy dots, fullscreen detection
Native code	C# compiled at runtime (Add-Type)	Spotify process-loopback audio capture, FFT, volume
Storage	node:sqlite (read-only), JSON	Notification database; %APPDATA%\islet\config.json
Registry	HKCU Notifications\Settings	Turning Windows' banners off and back on
Network	fetch()	lrclib.net, Open-Meteo, your ICS link, optional Spotify Web API
Agent hooks	HTTP server on 127.0.0.1:47321	Claude Code and Codex hooks post events
Tests	node --test	18 files, 148 tests
Packaging	electron-builder 25 (NSIS)	One-click per-user installer

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Key decisions

Decision	Why
No UI framework	One small view. Vanilla JS starts fast and is easy to read.
C# inside PowerShell	Smart App Control blocks unsigned exes. Add-Type compiles inside signed powershell.exe.
Read the notification DB	Windows has no notification-listener API for unpackaged apps.
Hooks over loopback HTTP	Instant agent status; transcript timing is the fallback.
powerMonitor for the charger	Plug / unplug events arrive instantly, no polling.
Windows media controls	No Spotify login or Premium needed; works with other players too.

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SECTION 05

Architecture

Electron's two-process model. The main process talks to Windows and the web through helpers; the notch page only sees a small, safe API.

Layer	Parts
Windows + web	Media session (SMTC) · Spotify audio (process loopback) · Registry (consent, banners) · Notification DB (wpndatabase.db) · Web APIs (lyrics, weather, ICS) · Downloads folder · Claude / Codex hooks · Power events
Helpers	media.ps1 · spotify-audio.ps1 + .cs · system.ps1 · notifications.js + icons.js · lyrics.js, weather.js, calendar.js, spotify-web.js · downloads.js · shelf-file.js · claude-bridge.js + hooks/islet-hook.js
Main process	main.js: window · tray · hotkeys · powerMonitor · config.json · IPC · starts every helper
Bridge	preload.js (window.bridge) · settings/preload.js (settings:load / save)
Screen	renderer/ (index.html · notch.css · notch.js · blackhole.js) · Settings window

How data moves

- **In:** each helper prints JSON lines (or POSTs, for hooks). main.js decodes them with one UTF-8 decoder per stream (the hook and the HTTP bridge do the same), parses them and forwards with webContents.send.
- **Draw:** notch.js keeps the latest state and calls render() on every event and every 500 ms.
- **Out:** clicks and scrolls call window.bridge (ipcRenderer.send). main.js runs them: writes to media.ps1, spotify-audio stdin, or answers the waiting hook request.
- **Mouse:** the window is click-through. Entering the notch sends hover(true); main.js makes it clickable until the pointer leaves.
- **Lifetime:** the notch window only closes when the app quits (tray Quit or Windows shutdown). Alt+F4 is ignored so helpers never talk to a destroyed window.

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SECTION 06

How each part works

Music	media.ps1 reads the Windows media session once a second and prints JSON; commands toggle, next, prev, seek and like come back on stdin. spotify-audio captures only the Spotify process tree into 24 log-spaced bands about 30 times a second. lyrics.js caches timed lyrics per song and resends the current ones when the page reloads. Spotify Web API answers that a newer request has overtaken are dropped, so Like acts on the right track.
Agents	claude-bridge.js runs an HTTP server on 127.0.0.1:47321 and keeps a status per session. hooks/islet-hook.js POSTs each event. A PermissionRequest is held for permissionWaitSeconds; with no answer the terminal asks as usual.
Notifications	notifications.js polls wpndatabase.db read-only every 2 s and parses the toast XML. Banners are hidden with ShowBanner = 0; old values are written to banners.json before the registry changes and kept until the restore is verified. On lock they are put back; on unlock hidden again.
Battery	main.js forwards powerMonitor on-ac / on-battery / resume. The renderer ignores the first report, dedupes, debounces 200 ms, and plays blackhole.js; an opposite event cancels it. Reduced motion shows a still frame.
System	system.ps1 reads the CapabilityAccessManager consent store for mic, camera and screen use, and detects fullscreen windows.
Info	weather.js (every 15 min). calendar.js: next timed event in 24 h; jumps straight to the current repeat, skips cancelled and excluded ones. Times follow TZID, including Outlook's Windows zone names; repeats keep their clock time in the event's own zone. DAILY and WEEKLY rules with INTERVAL, COUNT, UNTIL, BYDAY and WKST are followed; a rule with anything else, like BYMONTH, shows only its first date. downloads.js: .crdownload / .part / .partial, each finished file announced once.
Settings	Own window and preload. main.js validates every value, writes config.json via a temp file, and applies hand edits within 2 s. A broken config.json is left alone, not overwritten. A save's answer only refills fields you haven't changed since, so typing during an autosave is kept.

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SECTION 07

Repository map

Path	Lines	Purpose
main.js	653	Main process: window, tray, hotkeys, power events, config, IPC, starts every helper
preload.js	44	window.bridge: the only API the notch page can call
renderer/index.html	111	Notch markup: pill, pop-up, open view, panels
renderer/notch.css	289	Notch styling, states and animations
renderer/notch.js	572	Open / close, music, agents, battery, pop-ups, shelf, render loop
renderer/blackhole.js	224	Canvas black hole for charger in / out
settings/	177	Settings window: HTML, CSS, JS and its own preload
claude-bridge.js	192	Hook HTTP server; installs and removes Claude Code / Codex hooks
hooks/islet-hook.js	52	Run by each agent hook; POSTs the event
media.ps1	121	Windows media session reader and controller
spotify-audio.ps1 / .cs	325	Spotify loopback capture, FFT bands, volume
system.ps1	71	Mic / camera / screen dots, fullscreen detection
notifications.js	174	Notification database reader, banner toggle
icons.js	114	Real app logos
lyrics.js / weather.js	68 / 47	lrclib.net lyrics, Open-Meteo weather
calendar.js	209	Next event from an ICS link, time zones and repeats
downloads.js	77	Browser download progress
shelf-file.js	23	Reads and checks shelf.json
spotify-web.js	122	Optional Spotify Web API (PKCE login)
test/	1316	18 test files run by node --test
demo.html	191	Original browser design demo (not used by the app)
docs/ · package.json	- · 54	Screenshots · scripts, Electron version, installer config

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SECTION 08

Install and run

You need Windows 10 1903+ or 11, Node.js 20+ with npm, Git, and Windows PowerShell 5.1 (part of Windows).

Option A: run from the folder

```
git clone https://github.com/goprocker/Islet.git
cd Islet
npm install
npm start
```

Code changes apply on the next restart. Good for development.

Option B: install as an app

```
npm run dist      # creates dist\Islet Setup <version>.exe (run from PowerShell, not Git Bash)
```

Run the setup exe: it installs to %LOCALAPPDATA%\Programs\islet, adds Islet to the Start menu and starts it. The installed copy does not follow code edits; rebuild and reinstall to update. Quit any `npm start` copy first, as only one Islet runs at a time. The build is unsigned; if Smart App Control blocks it, use Option A.

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SECTION 09

Using Islet

Action	How
Open	Hover the notch
Close	Move away, or click anywhere else
Cycle pill / open / hidden	Ctrl+Alt+N
Next monitor	Ctrl+Alt+M, or tray > Show on screen
Volume / seek	Scroll over the music side / click the progress bar
Album view	Click the album art
Like	Heart button, or Alt+Shift+B
Clipboard / shelf / DND	Buttons in the tools column
Timer · Settings	Tray > Timer · Tray > Settings...
Quit	Tray > Quit (Alt+F4 does nothing on the notch)

SECTION 10

Claude Code and Codex

Tray > Connect to Claude Code and Codex installs hooks; untick to remove them. Backups are saved next to each file.

Hook event	What the notch shows
UserPromptSubmit, PostToolUse	Working, with the latest actions
Notification, PermissionRequest	Needs you; Claude Code requests get Allow / Deny
Stop, SessionEnd	Done
SessionStart (Codex)	New session chip

- Claude Code: ~/.claude/settings.json (backup settings.json.islet-backup).
- Codex: ~/.codex/hooks.json if ~/.codex exists (backup hooks.json.islet-backup). Approve Islet's hooks once in Codex (/hooks).
- Removing hooks deletes only Islet's handlers, even when they share a matcher group with your own.
- permissionWaitSeconds (default 8) is how long a request waits for you in the notch. 0 turns holding off.
- Codex permission requests show 'Needs you' but are answered in Codex.

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SECTION 11

Settings

Stored in %APPDATA%\islet\config.json and edited from the settings window.

Key	Default	Meaning
size	normal	compact, normal or large
accent	album	Colour from album art, or #rrggbb
idleHide	false	Thin line when nothing is happening
powerAnim NEW	true	Black hole animation when the charger goes in or out
powerAnimStyle NEW	cinematic	cinematic, or subtle (no particles or wave, softer glow, slower spin)
hideInFullscreen	true	Hide behind fullscreen apps
notifications	true	Show Windows notifications in the notch
hideWindowsBanners	true	Turn off Windows' own banners while running (back on at the lock screen)
dnd / dndFrom / dndTo	off	Do Not Disturb, optional daily HH:MM window
permissionWaitSeconds	8	Allow / Deny hold time, 0 to 24
lyrics	true	Fetch lyrics from lrclib.net
weatherCity / weatherUnit	empty / C	City name (empty turns weather off); C or F
calendarIcsUrl	empty	Private iCal link
downloadsFolder	empty	Empty means Downloads
clipboardHistory	true	Keep the last 5 copied texts in memory
displayId	null	Screen to show on
spotifyClientId	empty	Optional Spotify Web API (Premium)

If config.json can't be read (for example during a hand edit), Islet keeps the current settings and leaves the file as it is.

SECTION 12

Build, test, debug

```
npm test        # node --test test/*.test.js  (18 files, 148 tests)
npm run dist    # NSIS installer, run from PowerShell

# inspect the notch with Chrome DevTools
node_modules\electron\dist\electron.exe . --remote-debugging-port=9333
```

Open chrome://inspect to see the page. Handy globals: setOpen(true), render().

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SECTION 13

Privacy, troubleshooting, uninstall

Privacy

- Everything runs on your PC. Network calls only go to lrclib.net, Open-Meteo, your ICS link, and Spotify if connected.
- Clipboard history never touches disk. The shelf stores file paths only.
- The hook server listens on 127.0.0.1 only. The notch page has no direct Node access.
- Registry changes are limited to banner flags under HKCU and are put back on quit.

Troubleshooting

Problem	Fix
Banners still bottom-right	Start Islet normally (sandboxed shells can't write the registry). Sign out and in once if needed.
Banners off after a crash	Start Islet and quit from the tray, or turn the setting off.
Codex column empty	Approve Islet's hooks in Codex (/hooks).
npm run dist fails in Git Bash	Run it from PowerShell.
Installer blocked	Smart App Control; run with npm start.
Installed app ignores my code changes	Rebuild with npm run dist and run the new setup exe.
Second launch does nothing	Only one copy runs; it opens the settings of the running one.
Notch won't close with Alt+F4	By design; use tray > Quit.
Calendar event missing	Its time zone isn't recognised (skipped with a console warning). A repeat rule with parts like BYMONTH shows only its first date.
Shelf starts empty	shelf.json wasn't a valid list. The original is kept as shelf.json.invalid.

Uninstall

- Tray: untick Connect to Claude Code and Codex, and Start with Windows.
- Tray: Quit (puts Windows banners back).
- Installed app: Windows Settings > Apps > Islet > Uninstall. From source: delete the folder.
- Delete %APPDATA%\islet.

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SECTION 14

What's new in 2.0

Version 2.0 closes 14 reliability issues and collects everything added since the first guide. Nothing changes in how you use Islet, and no settings were added or renamed.

Fixed in 2.0

- **Calendar**: events follow their own time zone instead of the PC's; repeats keep their time across daylight saving; rules Islet doesn't fully understand no longer invent dates, so they show only their first date.
- **Notifications**: the banner recovery record is saved before banners are turned off; CDATA text displays correctly; pop-ups that arrive while hidden wait and resume instead of being lost.
- **Music**: a late Spotify reply can't make Like act on the wrong track; lyric lines with several timestamps show at every one; lyrics return after a page reload; a slow album cover can't override a fixed accent colour.
- **Settings**: typing during an autosave is kept, and a change made before the settings load no longer saves an empty form.
- **Shelf and downloads**: a damaged shelf.json no longer causes a startup error and is kept as shelf.json.invalid; downloads finishing together each announce their own file.
- **Agents**: emoji and other non-English text in prompts, folders and session names stay intact.
- **Tests**: 8 files, 43 passing → 18 files, 148 passing.

Also since the first guide

- **Black hole charger animation** with Cinematic / Subtle styles and an on/off switch (replaces the plain charging pop-ups).
- **Full notification text**: wider, taller pop-ups that stay longer for long messages; Windows banners return on the lock screen.
- **Alt+F4 fix**: the notch no longer closes and throws errors; quit from the tray.
- **Reliability fixes**: config.json never overwritten when broken, safe banner recovery, UTF-8 helper output, Calendar exceptions, shelf duplicates, waiting agent sessions kept, hook removal limited to Islet's own, wave loop stops when idle, charger glow vs. black hole / hide / idle conflicts.

Islet

Designed and built by **Gopinath**

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